

Valvular Heart Disease: Approach to a Case of Dyspnoea

Dr. Muhammad Reihan, Internal Medicine | Source: Approach to a case of Dyspnea – VHD (61 slides)

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 Bridge from earlier lectures:

- **Rheumatic fever** scars the mitral valve: early **MR**, late **MS**.
- **Infective endocarditis** causes **acute AR/MR** and colonises damaged or prosthetic valves.
- **CAD: papillary muscle rupture** → acute MR; ischaemic LV → **secondary (functional) MR**.
- **Pulmonary vascular disease:** MS → **pulmonary hypertension** (group 2 PH) → **functional TR**.
- **Pericardial:** constriction and TR both cause **ascites + raised JVP**.
- **Hypertension:** chronic AR gives a **wide pulse pressure**; AS gives a **narrow** one.

1. Title / Objectives

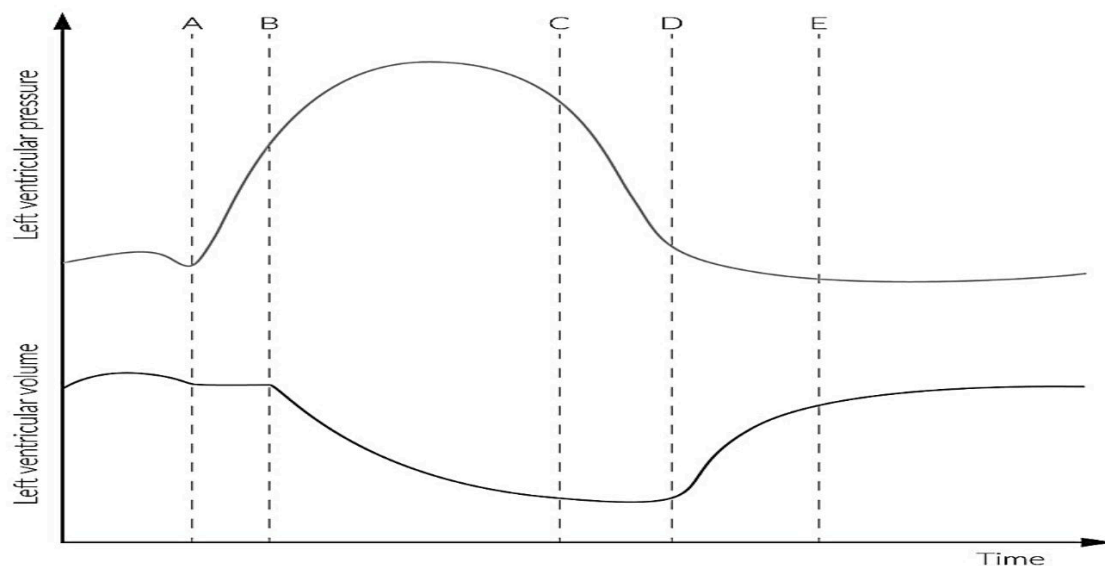
Domain	Objective	Likely tested as
Knowledge	★ Clinical features, auscultatory findings and complications of each major VHD	Murmur ↔ lesion; manoeuvres
Knowledge	Types, indications and thromboembolic risk factors of prosthetic valves	Mechanical vs bioprosthetic by age
Skill	★ Choose surgical vs medical management by severity, symptoms and LV function	LVEF cut-offs (AR ≤ 55%, MR ≤ 60%)
Skill	★ Urgent plan for acute severe regurgitation	Acute MR/AR → urgent surgery

Lecture flow: Opening case → Overview (definition, valve anatomy, epidemiology, auscultatory sites, stenosis vs regurgitation) → **AR** → **AS** → **MR** → **MVP** → **MS** → **TR** → Auscultation summary → **Prosthetic valves**.

2. Main Content (Lecture Order)

2.0 Opening Case (Slides 5-6)

- **Patient:** 75-year-old woman, **syncope during aerobic exercise** (10 s).
- **History:** 3 years of progressive **exertional dyspnoea + chest discomfort**.
- **Exam:** radial pulses **decreased and delayed bilaterally** (= *pulsus parvus et tardus*).
- **Question:** a plot of **LV pressure and LV volume** against time with points **A-E**. At which point does the murmur **start**? (Answered in §7, Q1.)



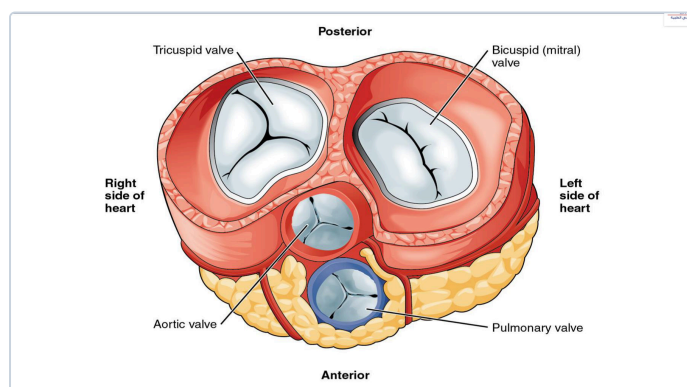
Slide 6: LV pressure and volume curves, points A-E

[added] Reading the curve:

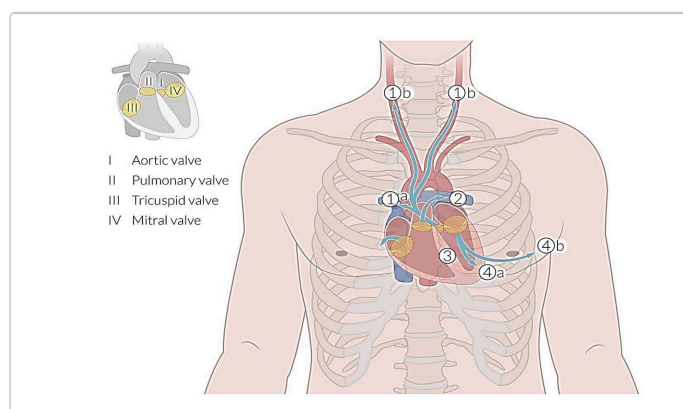
- **A** = mitral valve closes (S1; isovolumetric contraction starts; volume flat).
- **B** = **aortic valve opens** (volume starts to fall = ejection).
- **C** = aortic valve closes (S2; pressure falls steeply).
- **D** = mitral valve opens (volume rises = filling).
- **E** = mid/late diastole.

2.1 Definition & Anatomy (Slides 7-10)

VHD = a group of conditions affecting the heart valves. They are **acquired or congenital** and manifest as **stenosis and/or insufficiency (regurgitation)**.



Slide 8: the four valves seen from above



Slide 10: anatomical vs auscultatory sites

Epidemiology (slide 9):

Lesion	Fact
★ Aortic stenosis	Most common valve defect in industrialised countries ; mostly degenerative ; usually symptomatic after 75 . In young people it is usually congenital (bicuspid valve)

Lesion	Fact
Mitral stenosis	Symptom onset 20-39 years (rheumatic)
Mitral regurgitation	Second most common valve defect

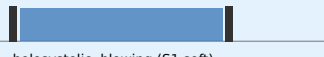
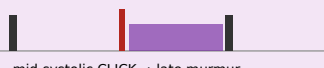
Auscultation areas *[added, standard]:*

- **Aortic:** right 2nd ICS
- **Pulmonary:** left 2nd ICS
- **Tricuspid:** left lower sternal edge
- **Mitral:** apex (5th ICS, mid-clavicular line)

2.2 Effects of Stenosis vs Regurgitation (Slide 11) ★

	Stenosis	Regurgitation
Load	Pressure overload on the chamber behind the valve	Volume overload: the heart pumps the same blood twice
Adaptation	Hypertrophy (concentric <i>[added]</i>) → dilatation	Dilatation (eccentric hypertrophy <i>[added]</i>)
End result	Heart failure	Heart failure

Murmur map: timing → shape → site → radiation → manoeuvre

Lesion	Shape (S1 ... S2 ... S1)	Where / radiates	Extra sounds	Louder with
AS systolic	 S1 · click · diamond · soft S2	R 2nd ICS → both carotids Pulsus parvus et tardus	Ejection click, S4 Soft/single S2	↑ preload (squat, leg raise) ↓ handgrip, Valsalva
MR systolic	 holosystolic, blowing (S1 soft)	Apex → LEFT AXILLA Displaced apex (LV dilated)	S3 (volume load)	Handgrip (↑ afterload) Leg raise (↑ preload)
MVP systolic	 mid-systolic CLICK → late murmur	Apex (± axilla) Young women, Marfan, EDS	Midsystolic click	Standing / Valsalva (click earlier, longer) ↓ squatting
AR diastolic	 S2 → early decrescendo, blowing	L sternal edge (3rd-4th ICS) Sit up, lean forward, expiration	S3, Austin Flint (apical mid-diastolic)	Handgrip (↑ afterload) Collapsing pulse, wide PP
MS diastolic	 LOUD S1 · S2 · opening snap · rumble	APEX, left lateral position, bell No radiation	Loud S1, opening snap (short S2-OS = severe)	Exercise, left lateral Malar flush, AF
TR systolic	 pansystolic (like MR)	L lower sternal edge → apex, NOT to axilla	Giant v waves, pulsatile liver	INSPIRATION (Carvallo) Right-sided ↑ on inspiration

Rule: right-sided murmurs ↑ on inspiration; left-sided ↑ on expiration. Only HCM and MVP get LOUDER with Valsalva/standing.

Added diagram: murmur map (timing, shape, site, radiation, manoeuvres)

🔑 **Recap: Overview.** Stenosis = pressure load → hypertrophy; regurgitation = volume load → dilatation; both end in HF. AS is the commonest lesion in industrialised countries (degenerative, > 75; bicuspid if young). MR is second. MS presents at 20-39 years (rheumatic).

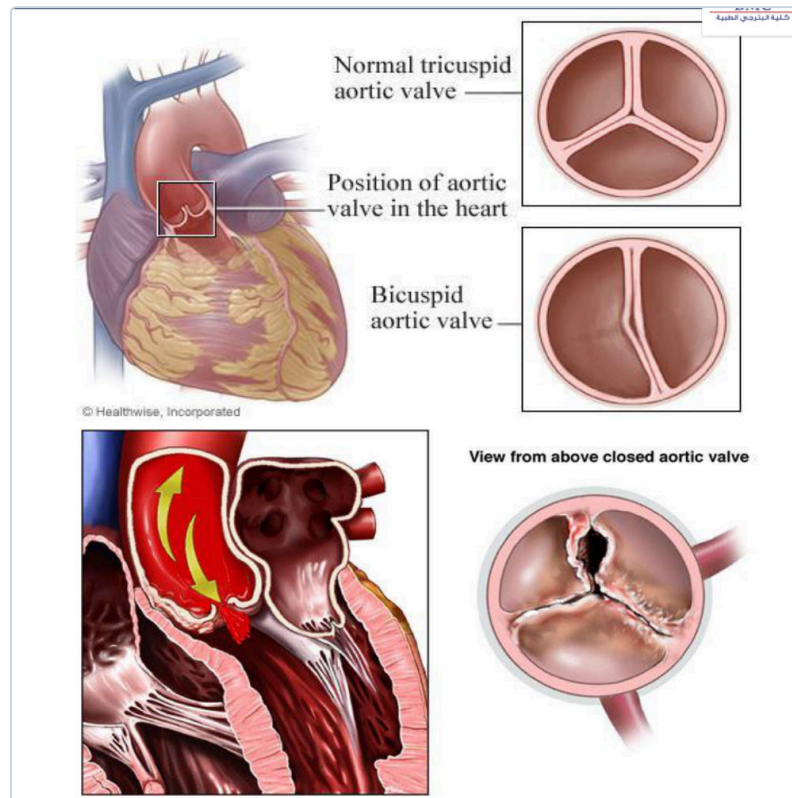
🔗 **Link:** The lecture starts with the aortic valve, beginning with regurgitation.

👉 **Up next: Aortic regurgitation.**

2.3 Aortic Regurgitation (Slides 12-23) ★

Causes:

Acute AR	Chronic AR: valve	Chronic AR: root
★ Infective endocarditis	★ Congenital bicuspid AV : most common cause in young adults, high-income countries	Aortic dilatation
★ Aortic dissection	★ Rheumatic heart disease : most common cause in lower-income countries	Connective tissue disease: Marfan, Ehlers-Danlos
Chest trauma		Tertiary syphilis



Slide 14: bicuspid vs tricuspid aortic valve

Clinical features:

- ★ **Acute AR: sudden severe dyspnoea, rapid decompensation, pulmonary oedema.** The LV has no time to dilate, so there are *[added]* **no** peripheral signs.
- Chronic AR: may be asymptomatic for decades** despite progressive LV dilatation. - **Palpitations.** - **High pulse pressure signs:** water-hammer pulse, Traube, Quincke. - **Left HF:** exertional dyspnoea, **angina**, orthopnoea, fatigue, **syncope**.

Peripheral signs of AR (slide 18) ★

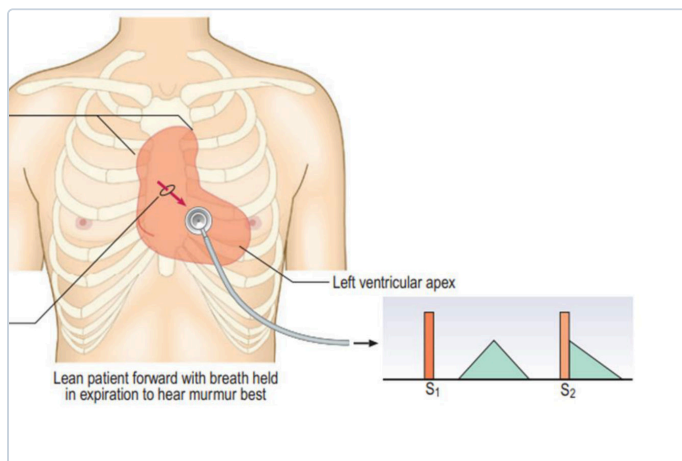
Sign	Finding
De Musset	Head bobbing with each heartbeat
Becker	Visible retinal arteriole pulsation
Müller	Pulsating uvula
Quincke	Capillary pulsation in the nail bed on light compression
Corrigan pulse	Rapid forceful distension with quick collapse (= water-hammer pulse)
Traube	"Pistol-shot" sounds over the femoral artery
Duroziez	Gradual femoral compression → systolic AND diastolic bruit

Sign	Finding
★ Hill	Popliteal SBP exceeds brachial SBP by ≥ 60 mmHg → most sensitive sign of AR

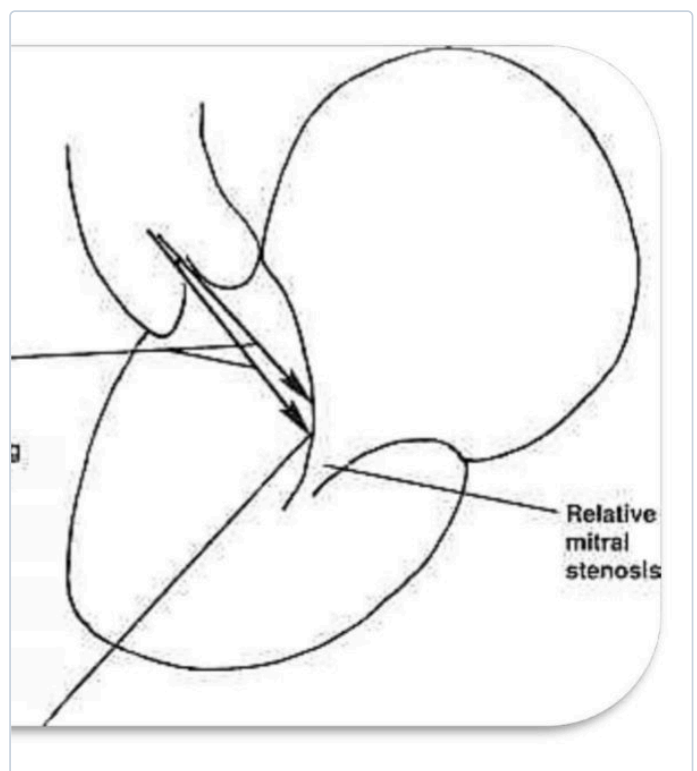
Mnemonic: the signs move **head → eye → mouth → nail → pulse → groin → knee**: De Musset · Becker · Müller · Quincke · Corrigan · Traube · Duroziez · Hill.

Auscultation (slides 19-20):

- **S3**
- ★ **High-pitched, blowing, DECRESCENDO EARLY DIASTOLIC murmur**, best heard at the **left sternal edge with the patient sitting up, leaning forward, breath held in expiration** (slide 19 figure)
- ★ **Austin Flint murmur**: *[added]* apical mid-diastolic rumble; the regurgitant jet stops the anterior mitral leaflet opening fully ("relative mitral stenosis", slide 20 figure)
- Severe stages: a harsh **crescendo-decrescendo mid-systolic** flow murmur



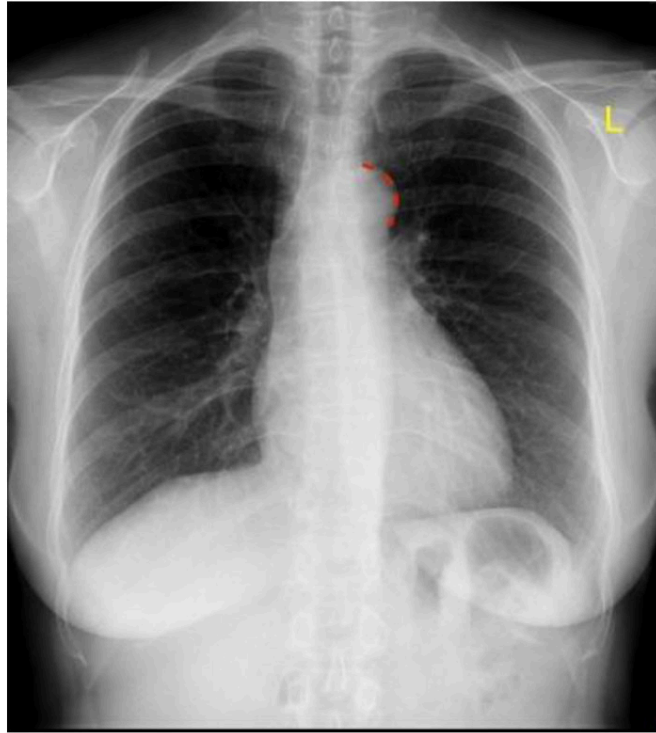
Slide 19: AR auscultation position



Slide 20: Austin Flint mechanism

Diagnostics (slides 21-22):

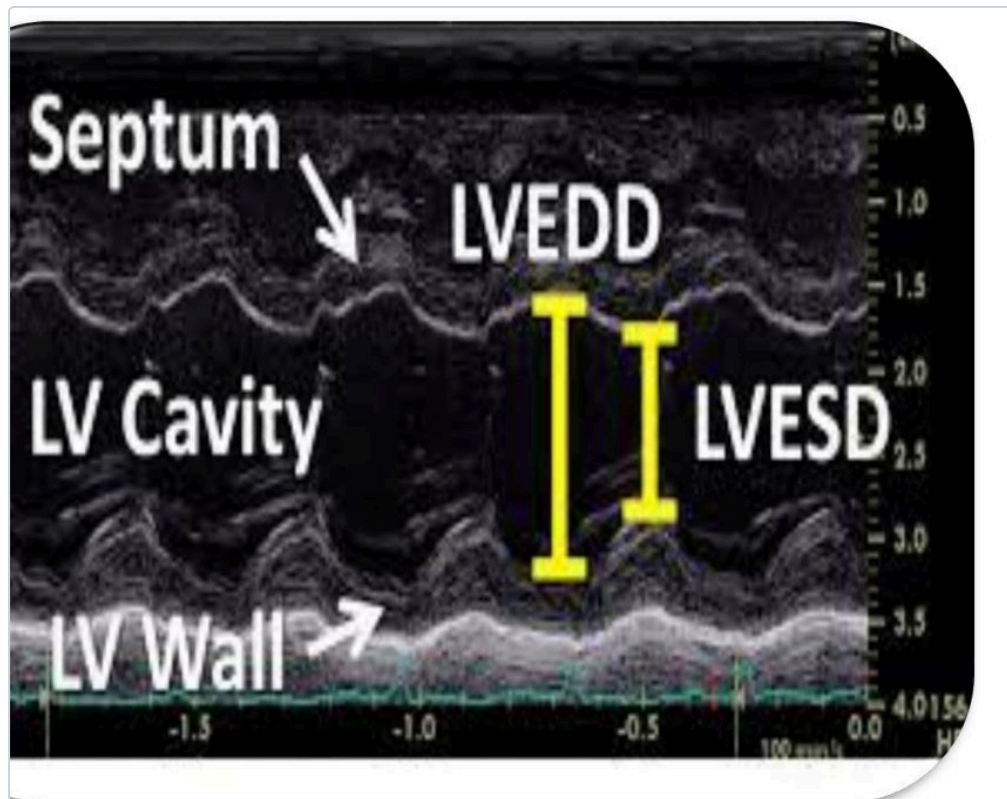
- **Echocardiography, TTE = modality of choice**: valve structure and function, cause, **severity**, LV, other valves.
- **CXR**: pulmonary oedema / other causes of dyspnoea. Chronic AR: **enlarged cardiac silhouette + prominent aortic root/arch**.



Slide 22: chronic AR on CXR

Treatment (slide 23) ★

Surgical indications (AVR)	Medical
Acute severe AR	Treat heart failure ([added] vasodilators: ACEI/ARB, DHP-CCB; avoid β -blockers in acute AR)
Symptomatic chronic severe AR	
Asymptomatic chronic severe AR with LVEF $\leq$ 55% , or consider if LVESD $>$ 50 mm	



Slide 23: LV dimensions (LVEDD / LVESD) on M-mode

🔗 **Recap: AR.** Acute AR (IE, dissection, trauma) → flash pulmonary oedema → urgent surgery. Chronic AR (bicuspid/rheumatic valve, or dilated root from Marfan/EDS/syphilis) is silent for years. Signs: wide pulse pressure, collapsing pulse, De Musset ... Hill (most sensitive). Early diastolic decrescendo murmur at the LSE leaning forward; Austin Flint at the apex. Operate if symptomatic, or if asymptomatic with $EF \leq 55\%$ / $LVESD > 50\text{ mm}$.

🔗 **Link:** Same valve, opposite problem: obstruction to outflow.

📌 **Up next: Aortic stenosis.**

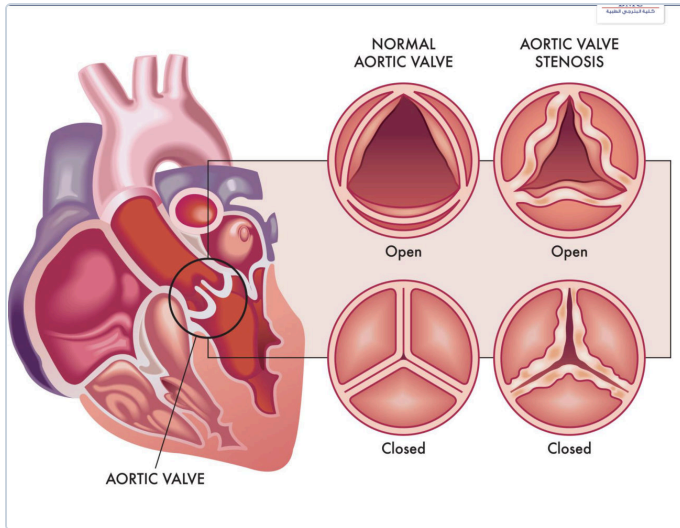
2.4 Aortic Stenosis (Slides 24-32) ★

Etiology (slides 25-27):

Cause	Detail
★ Aortic valve sclerosis / calcification	Most common cause ; prevalence ↑ with age
★ Bicuspid aortic valve (BAV)	Two of three leaflets fused in utero → symptoms EARLIER than senile calcification
Rheumatic fever	[added] almost always with mitral disease

BAV classification (slide 27, Sievers):

- **Type 0** (no raphe, "true" BAV): **6%**
- **Type 1** (1 raphe): **89%** (1a R-L 71%, 1b R-N 15%, 1c L-N 3%)
- **Type 2** (2 raphes, functionally unicuspid): **5%**



Slide 25: normal vs stenotic aortic valve

Classification of BAV		
Type 0 (0 raphe, true BAV)	Type 1 (1 raphe)	Type 2 (2 raphe)
 (6%)  (4%) lat  (2%) AP	 (89%)  Type 1a (71%) R-L  Type 1b (15%) R-N  Type 1c (3%) L-N	 (5%) (functionally unicuspid)

adapted from Sievers HH, Schmidtke C. J Thorac Cardiovasc Surg 2007; 133:1226-1233

Slide 27: BAV classification

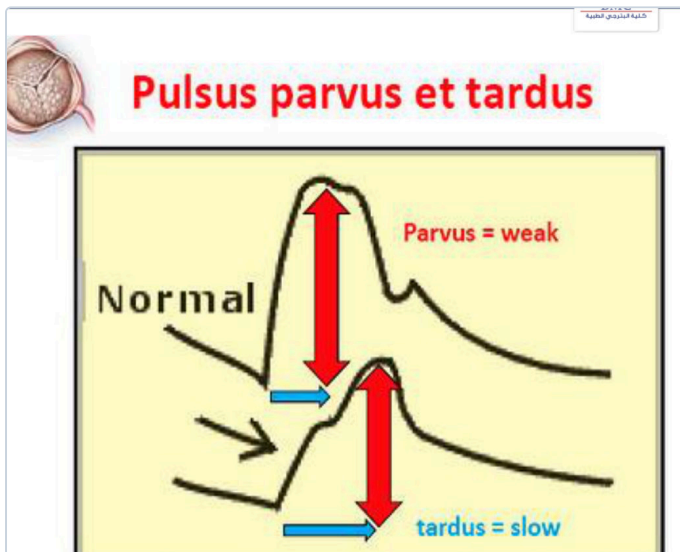
Clinical features (slide 28): may be **asymptomatic for years**.

★ **Mnemonic: SAD** = **S**yncope (and dizziness) · **A**ngina · **D**yspnoea (HF).

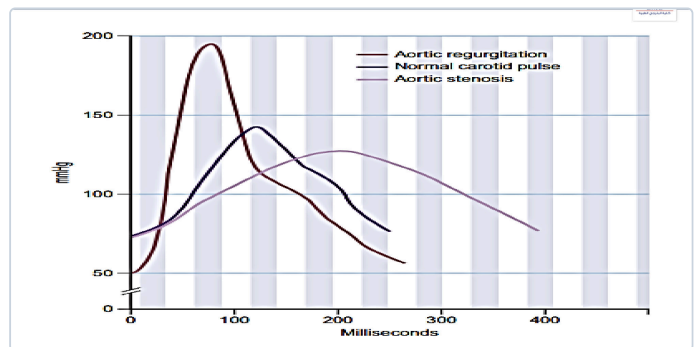
[added] Untreated survival after symptom onset: angina ~5 y, syncope ~3 y, HF ~2 y.

Examination:

- **Small BP amplitude / narrow pulse pressure.**
- ★ **Pulsus parvus et tardus:** weak (parvus) and delayed (tardus) carotid/distal pulse.



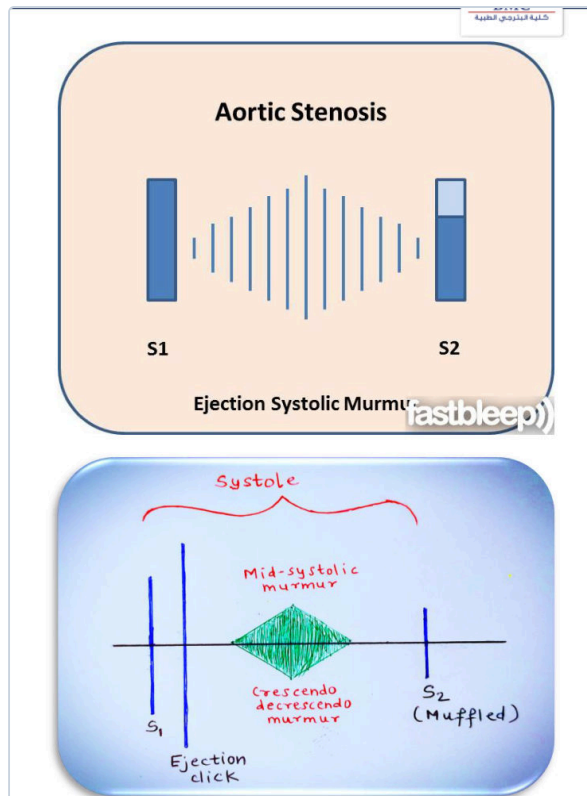
Slide 28: pulsus parvus et tardus



Slide 29: pulse waveforms (AR vs normal vs AS)

Auscultation (slide 30) ★

- **Harsh crescendo-decrescendo ("diamond") systolic ejection murmur**, radiating **bilaterally to the carotids**.
- **Handgrip** → **DECREASES** the murmur (↑ afterload → less gradient).
- **Valsalva and standing from squatting** → **decrease** it, or leave it unchanged (↓ preload).
- **Soft S2** ([added] single/paradoxically split when severe).
- **S4**, best heard at the apex.
- **Early systolic ejection click** (bicuspid valve).



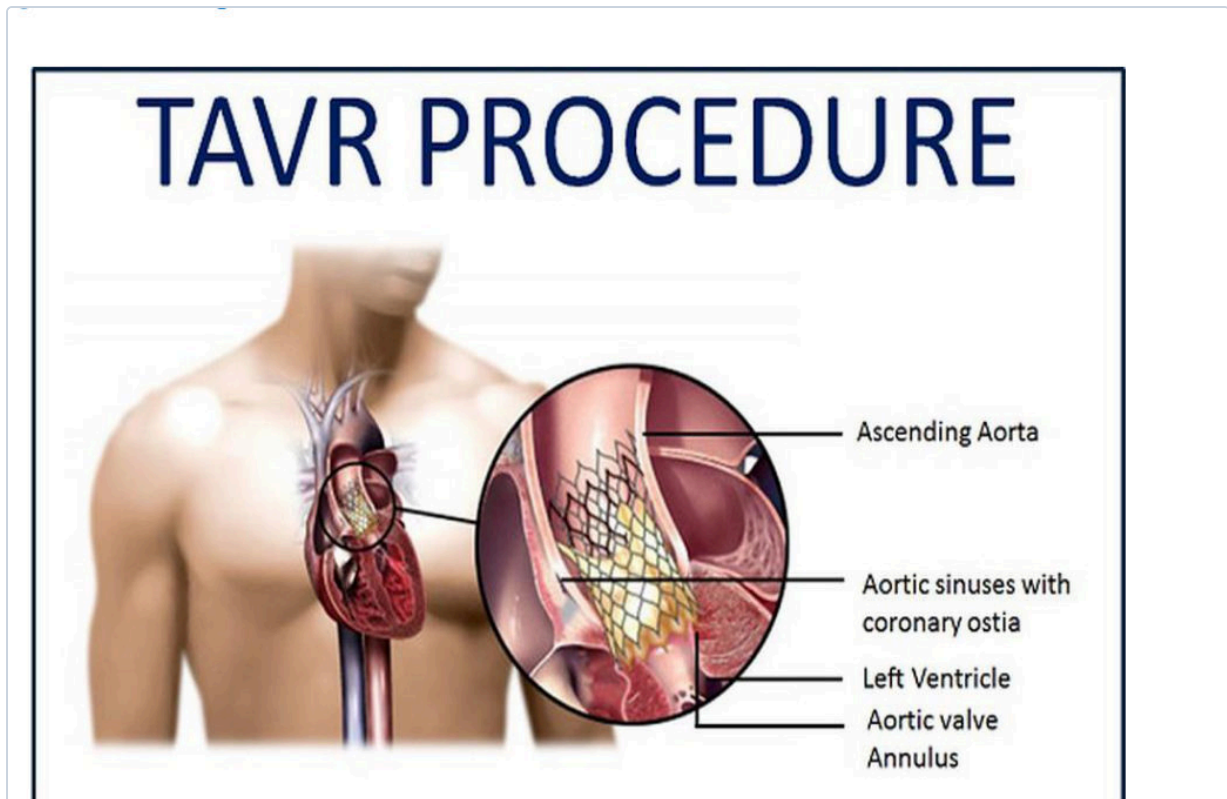
Slide 30: AS murmur (diamond-shaped ejection murmur)

Diagnostics (slide 31):

- **Echocardiography: primary test**; confirms the diagnosis and grades severity ([added] severe = **AVA < 1.0 cm², mean gradient ≥ 40 mmHg, Vmax ≥ 4 m/s**).
- **CXR**: pulmonary oedema, other causes of dyspnoea, **valve calcification** (more severe disease), **LVH**.
- **Cardiac catheterisation: accurate** measurement of **valve area**, cardiac output and **mean aortic gradient** ([added] used when echo is discordant).

Treatment (slide 32) ★

AVR indications	Procedures
★ ALL symptomatic patients	Surgical AVR (SAVR)
Asymptomatic severe AS with significantly ↓ LVEF ([added] < 50%)	★ Transcatheter AVR (TAVR) : [added] elderly / high surgical risk
Asymptomatic severe AS undergoing other cardiac surgery (e.g., CABG)	Percutaneous balloon valvuloplasty ([added] bridge / palliation; restenosis)



Slide 32: TAVR procedure

[added] Avoid vasodilators/nitrates in severe AS: with a fixed outflow, preload drop → syncope/shock. No medical therapy changes the natural history.**

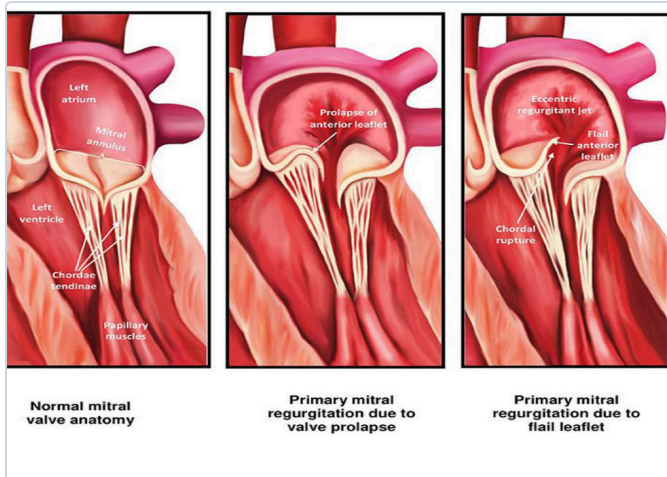
🔄 **Recap: AS.** The commonest valve lesion: senile calcific (> 75), bicuspid (younger), rheumatic. SAD = syncope, angina, dyspnoea. Pulsus parvus et tardus, narrow pulse pressure, ejection click, diamond murmur to the carotids, soft S2, S4. Handgrip and Valsalva soften it. Echo grades severity; cath gives the exact area. AVR (SAVR/TAVR) for any symptoms, low EF or concurrent surgery.

🔗 **Link:** Moving to the mitral valve, the regurgitant lesion comes first.

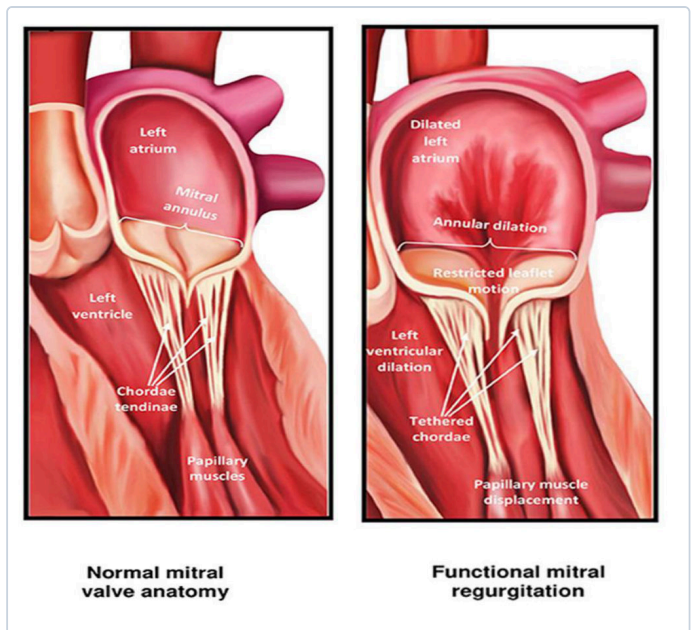
▶ **Up next: Mitral regurgitation.**

2.5 Mitral Regurgitation (Slides 33-40) ★

Primary (organic): leaflets/chordae	Secondary (functional): LV problem
Degenerative: ★ MVP, mitral annular calcification, ruptured chordae	★ CAD / prior MI affecting the papillary muscles
Rheumatic fever	Dilated cardiomyopathy (annular dilatation)
Infective endocarditis	



Slide 34: primary MR (prolapse, flail leaflet)



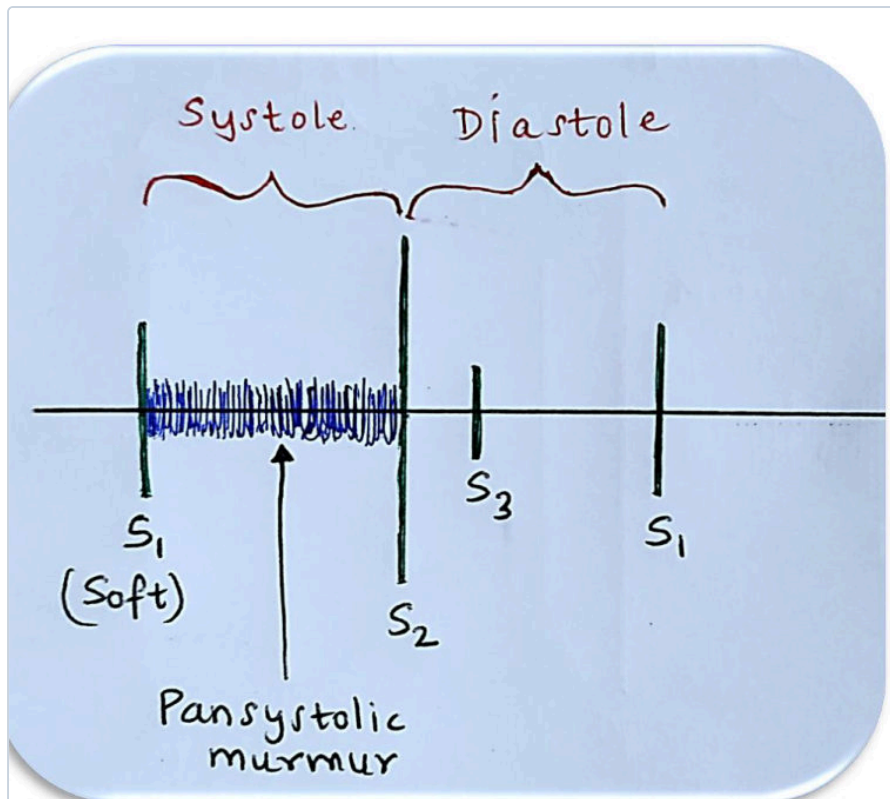
Slide 35: functional MR from a dilated LV

Clinical features (slide 36):

- **Acute MR:** left HF, pulmonary oedema, cardiogenic shock ([added] papillary rupture after MI, IE, chordal rupture).
- **Chronic MR:** dyspnoea (exertional), **dry cough**, **palpitations** ([added] AF), left HF.

Auscultation (slide 37) ★

- **S3** in advanced disease
- ★ **Holosystolic (pansystolic), high-pitched, blowing** murmur, **best at the apex**, radiating to the **LEFT AXILLA**
- **Louder with ↑ preload** (leg raise) and **↑ afterload (handgrip)** (more regurgitation)
- [added] **Soft S1**



Slide 37: MR pansystolic murmur

Diagnostics (slide 38):

- **Echo (TTE = modality of choice):** valve apparatus, LV/LA size and function, **grade MR**.
- **ECG** (remodelling): **LVH**, ★ **P mitrale** ([added] AF).
- **CXR:** pulmonary congestion, **LV and LA enlargement**.

Treatment (slides 39-40) ★

★ Acute primary MR	Urgent surgical repair or replacement for ALL. Meanwhile manage HF medically (diuretics, nitrates, afterload reduction ; [added] ± IABP)
Procedure	★ Valve REPAIR is preferred over replacement
Chronic severe primary MR: surgery or transcatheter mitral repair ([added] MitraClip)	Symptomatic (irrespective of LV function) · Asymptomatic with LV dysfunction, LVEF ≤ 60% ([added] or LVESD ≥ 40 mm)

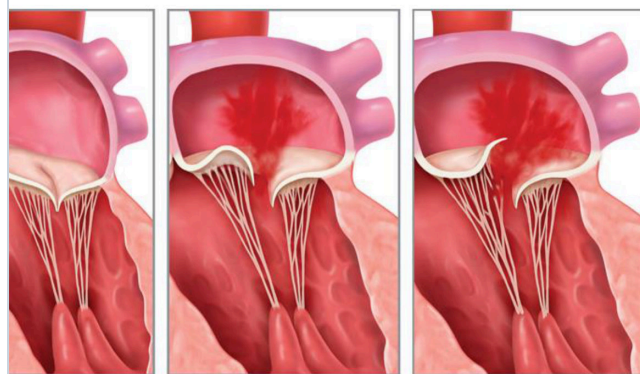
★ **EF cut-offs: AR ≤ 55% · MR ≤ 60%.** In MR, the EF is flattered by ejection into the low-pressure LA, so a "normal-looking" 60% already means dysfunction [added].

2.6 Mitral Valve Prolapse (Slides 41-45)

- **Epidemiology:** ★ **the most common cause of MR in developed countries.**
- **Etiology:** mostly **idiopathic**; **Marfan, Ehlers-Danlos**; acute rheumatic heart disease; IE.
- **Pathology:** ★ **myxomatous degeneration** → prolapse → **flail leaflet**.
- **Clinical:** - **Most patients are asymptomatic.** - ★ **Mid-systolic click** (high-frequency, mitral area) → **mid-to-late systolic murmur**, radiating to the axilla. - **Squatting diminishes the murmur**; [added] standing/Valsalva bring the click **earlier** and lengthen the murmur (slide 44 figure).
- **Diagnosis:** **TTE (test of choice).**

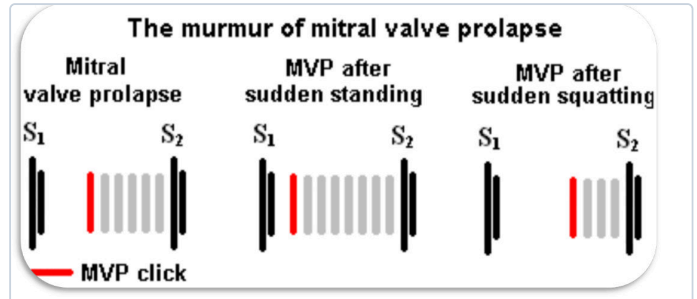
- **Treatment: none in most cases.**
- **Complication: IE** ([added] plus severe MR, arrhythmia, rarely sudden death).

ous degeneration



Normal Valve Degenerative MR—Prolapse Degenerative MR—Flail

Slide 43: myxomatous MVP and flail leaflet



Slide 44: MVP click and murmur with standing / squatting

🔄 Recap: MR and MVP.

- **MR:** primary (MVP, annular calcification, chordal rupture, RF, IE) vs secondary (ischaemic papillary dysfunction, DCM). Acute MR → pulmonary oedema/shock → urgent surgery (repair preferred). Chronic MR → pansystolic apical murmur to the axilla, S3, P mitrale, LA/LV enlargement. Operate if symptomatic or EF ≤ 60%.
- **MVP:** myxomatous; the commonest MR cause in developed countries; click then late systolic murmur; squatting softens it; usually no treatment.

🔗 **Link:** The other rheumatic mitral lesion is the late, scarred one: stenosis.

👉 **Up next: Mitral stenosis.**

2.7 Mitral Stenosis (Slides 46-53) ★

- **Etiology:** ★ **most commonly rheumatic fever.**

Clinical features (slide 47):

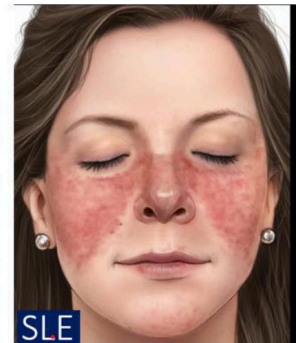
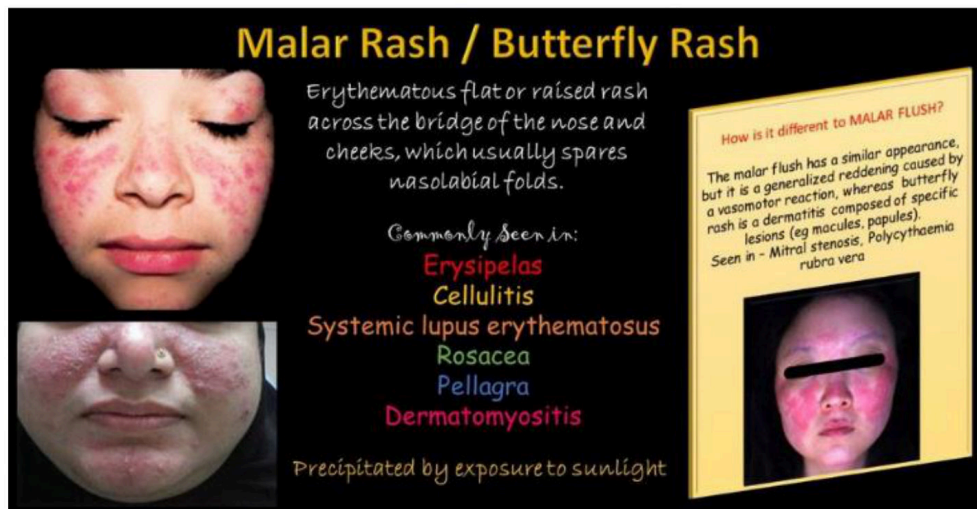
- **Dyspnoea**
- ★ **Hoarseness:** [added] **Ortner syndrome**; a big LA compresses the left recurrent laryngeal nerve
- ★ **Dysphagia:** [added] LA compresses the oesophagus
- **Later stages:** - **right HF symptoms** - **PND** - **orthopnoea** - ★ **haemoptysis** ([added] bronchial vein rupture)

Malar flush vs malar rash (slide 48):

- ★ **Malar flush (mitral facies):** **pinkish-purple patches** on the cheeks + **peripheral cyanosis** (lips, nose tip, cheeks). [added] Low cardiac output + vasoconstriction.
- **SLE butterfly rash:** erythematous, flat or raised, **spares the nasolabial folds**, photosensitive.

Malar flush (mitral facies):

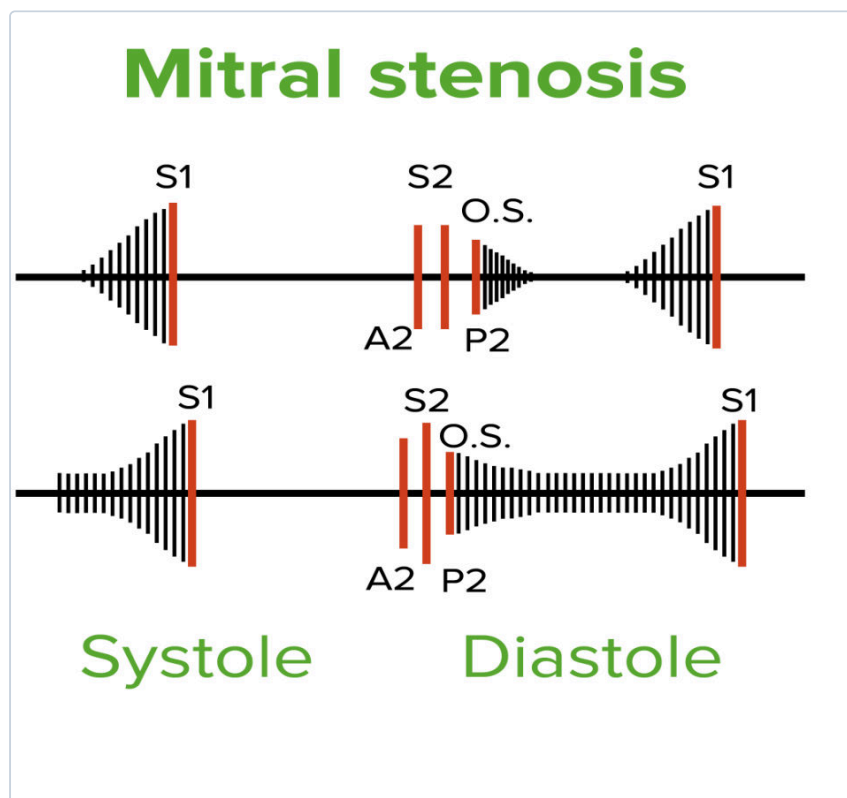
pinkish purple patches on cheeks, peripheral cyanosis in lips, tip of nose and cheek.



Slide 48: malar flush (MS) vs malar rash (SLE)

Auscultation (slide 49) ★

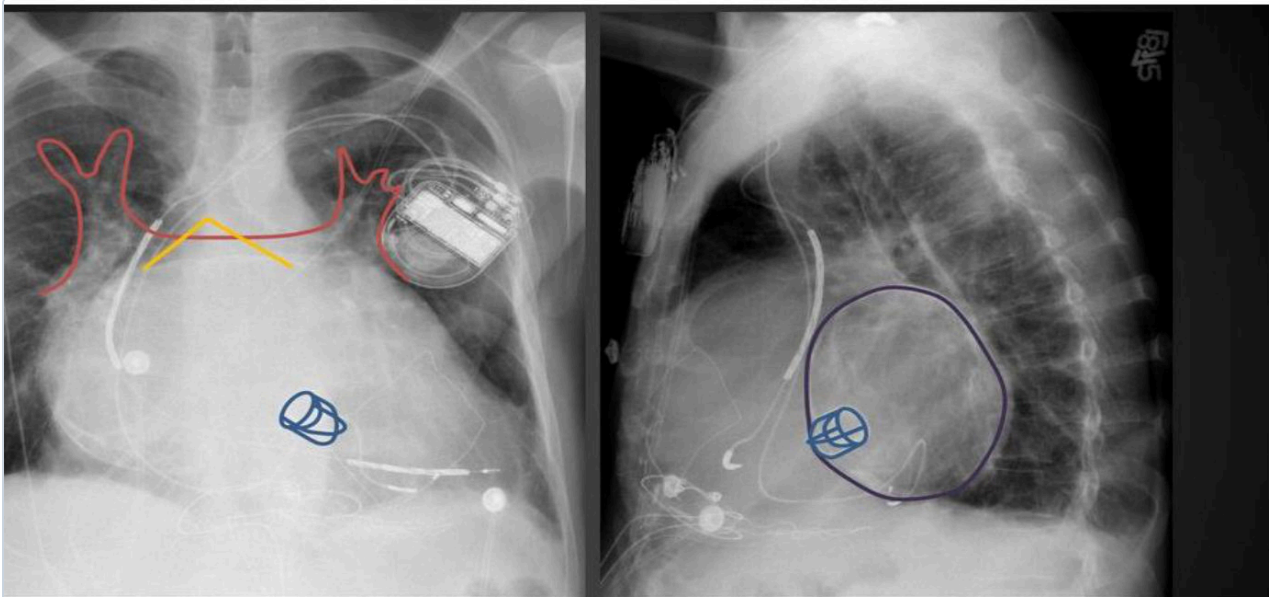
- **Low-pitched diastolic rumbling murmur, best at the apex** ([added] bell, left lateral decubitus)
- ★ **Loud S1**
- ★ **Opening snap.** [added] **The shorter the S2-OS interval, the more severe the MS** (slide 49 diagram: severe = OS closer to S2, longer rumble)
- [added] **Pre-systolic accentuation** (lost in AF)



Slide 49: MS murmur, mild vs severe

Diagnostics (slides 50-51):

- **TTE = most important** for diagnosis and guiding treatment: - **↓ mitral valve area** ([added] severe $\leq 1.5 \text{ cm}^2$) - **Thickened, calcified leaflets with commissural fusion** ([added] "hockey-stick" anterior leaflet) - **LA enlargement**
- **ECG: LA enlargement = P mitrale** ([added] AF).
- **CXR (PA + lateral):** - **LA enlargement** - **Enlarged LA appendage** ([added] straight left heart border, double right-heart contour, splayed carina) - **Pulmonary congestion**



Slide 51: LA enlargement on CXR

Treatment (slide 52):

- **Acute stabilisation:** standard HF therapy (**diuretics**; [added] **rate control** in AF, since a slower rate allows LV filling; **anticoagulate if AF**).
- ★ **Percutaneous mitral balloon commissurotomy (PMBC): preferred intervention** for most severe MS ([added] if the valve is pliable, no LA thrombus, $\leq$ moderate MR).
- **Surgery: open commissurotomy or mitral valve replacement** (mechanical or bioprosthetic).

Complications (slide 53):

- ★ **Atrial fibrillation** ([added] → LA thrombus → **stroke**)
- **Pulmonary oedema**
- **Pulmonary hypertension**
- **CHF**

🔗 **Recap: MS.** Rheumatic, presenting at 20–39 years. Dyspnoea, haemoptysis, Ortnier hoarseness, dysphagia, malar flush. Loud S1, opening snap, apical mid-diastolic rumble (short S2–OS = severe). P mitrale, AF, LA enlargement. PMBC is first-line intervention; surgery (commissurotomy/MVR) otherwise. Complications: AF/stroke, pulmonary oedema, pulmonary HTN, HF.

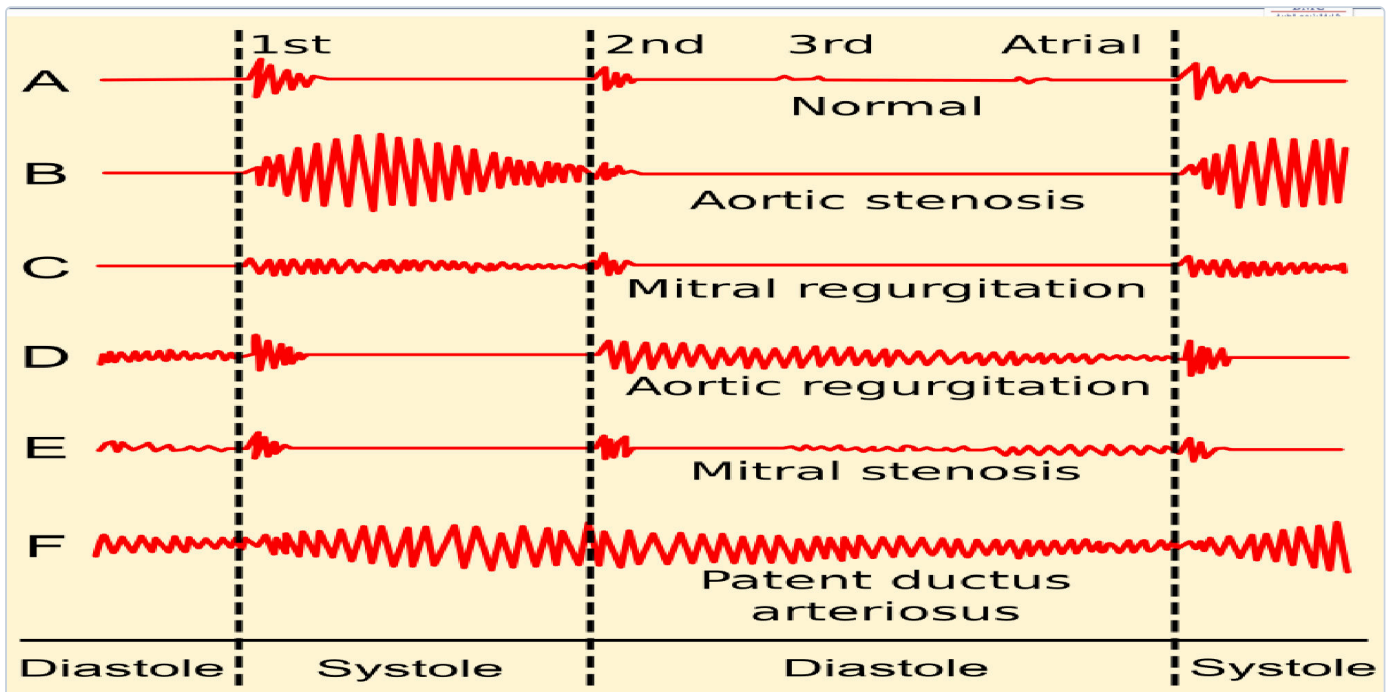
🔗 **Link:** Pulmonary hypertension from left-sided disease dilates the RV, which stretches the tricuspid ring.

▶ **Up next: Tricuspid regurgitation.**

2.8 Tricuspid Regurgitation (Slide 54)

- **Aetiology:** ★ **usually FUNCTIONAL:** RV enlargement → **dilatation of the tricuspid ring** ([added] from pulmonary HTN, left-heart disease or MS). Primary causes [added]: IE (IVDU), carcinoid, Ebstein, pacemaker leads, rheumatic disease.
- **Haemodynamics:** in systole, blood regurgitates from RV → RA → **↑ RA pressure** → RA enlargement + **systemic congestion** + **low cardiac output (LCOP)**.
- **Clinical picture = RV failure:** - **Congested, pulsating neck veins with systolic expansion** ([added] giant cv waves) - **Enlarged, tender, pulsating liver** with mild jaundice - ★ **Ascites BEFORE lower-limb oedema** - **LCOP:** cold hands, weak pulse, low SBP, peripheral cyanosis
- **Auscultation:** ★ **pansystolic murmur INCREASED BY INSPIRATION** ([added] **Carvallo sign**). Maximal over the **tricuspid area, propagated to the apex but NOT the axilla**.

2.9 Cardiac Auscultation Summary (Slide 55)



Slide 55: phonocardiogram of common murmurs (normal, AS, MR, AR, MS, PDA)

Trace	Pattern
A. Normal	S1, S2 only
B. AS	Crescendo-decrescendo systolic (diamond)
C. MR	Holosystolic plateau
D. AR	Early diastolic decrescendo
E. MS	Mid-diastolic rumble with presystolic accentuation
F. PDA	Continuous "machinery" murmur through S2

🔄 **Recap: TR and auscultation.** TR is mostly functional (dilated RV ring). Raised JVP with systolic waves, pulsatile liver, ascites before oedema. Pansystolic murmur ↑ on inspiration, radiating to the apex but not the axilla. **Systolic:** AS (diamond), MR/TR (holosystolic), MVP (click + late). **Diastolic:** AR (early), MS (mid + snap). **Continuous:** PDA.

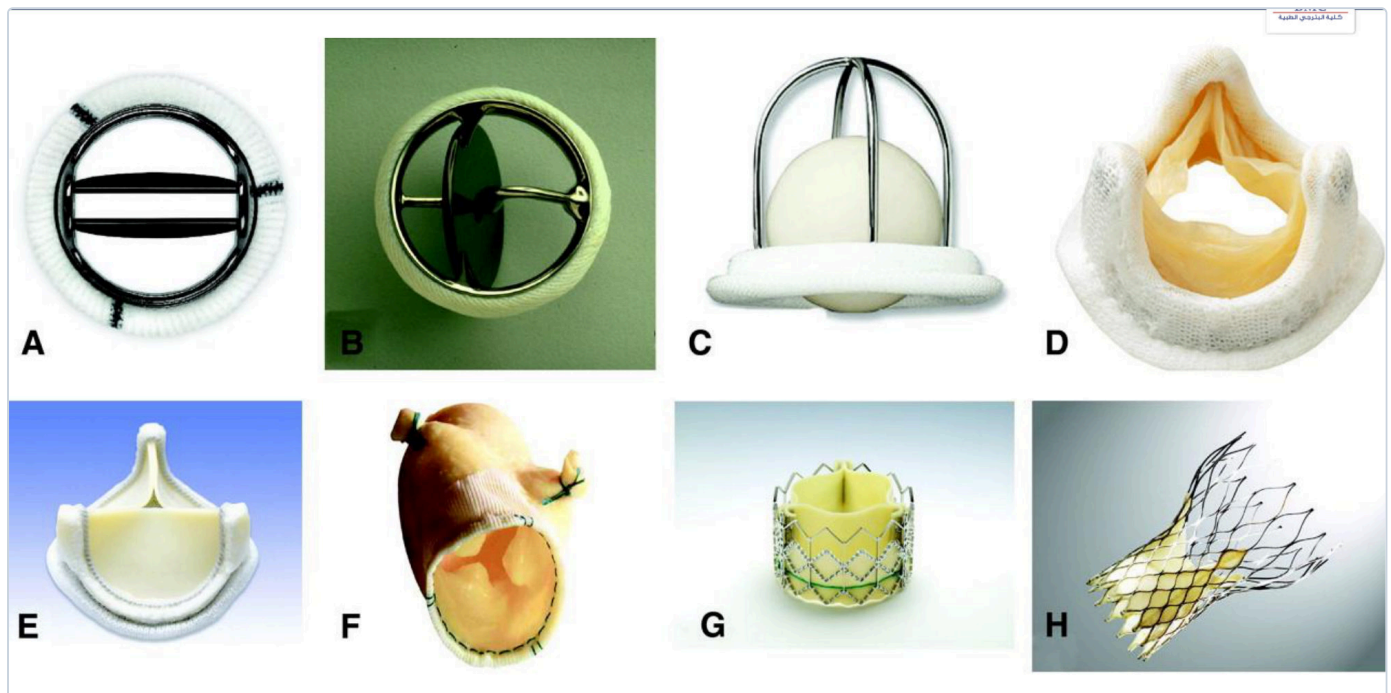
🔗 **Link:** When a valve must be replaced, the choice of prosthesis decides lifelong anticoagulation.

▶ **Up next: Prosthetic valves.**

2.10 Prosthetic Valves (Slides 56-59) ★

Types (slide 57). Identification [added, reconstructed from the standard figure]:

Letter	Probable type
A	Bileaflet mechanical (St Jude)
B	Tilting-disc mechanical
C	Caged-ball (Starr-Edwards)
D	Stented porcine bioprosthesis
E	Stented pericardial bioprosthesis
F	Stentless / homograft
G, H	Transcatheter (TAVR) valves: balloon-expandable and self-expanding



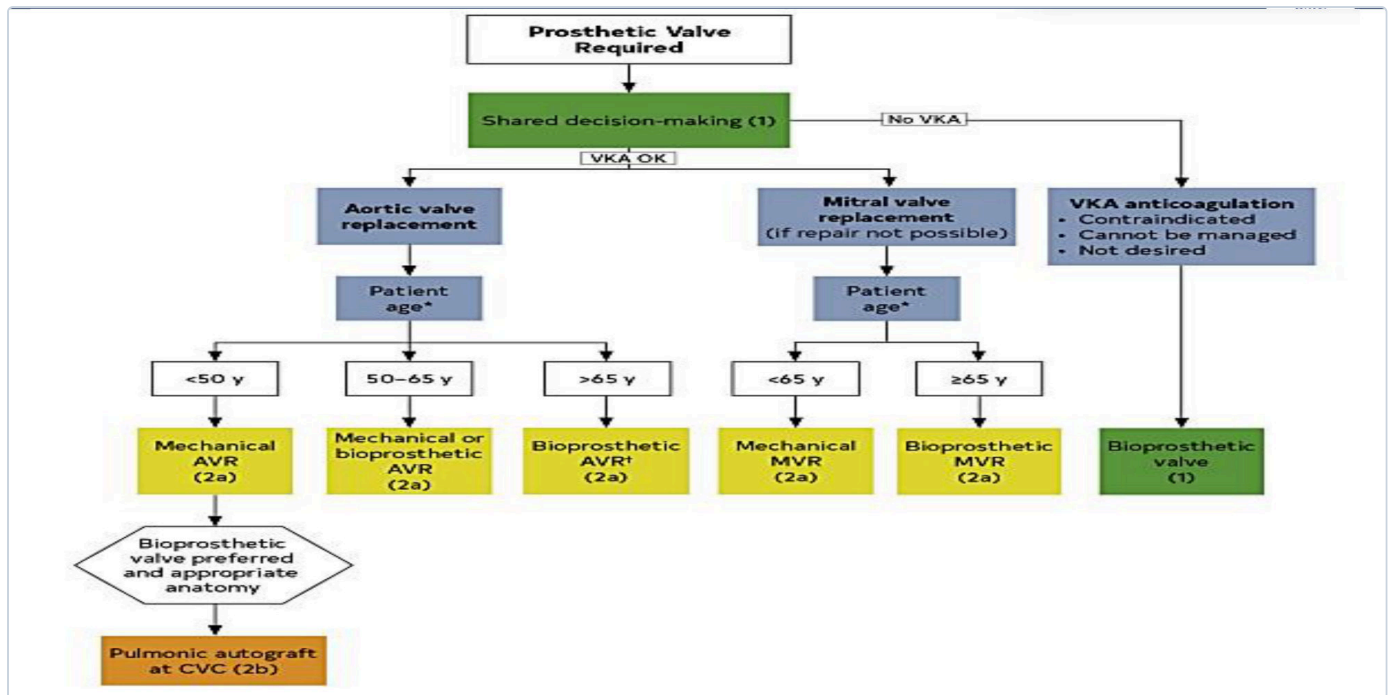
Slide 57: types of prosthetic valves

Choosing the valve (slide 58, ACC/AHA 2020) ★

- **Shared decision-making** first.
- **VKA possible:**

Position	Age	Recommendation
Aortic	< 50 y	Mechanical AVR (2a)
		If a bioprosthesis is preferred and anatomy suits → pulmonic autograft (Ross procedure) at a comprehensive valve centre (2b)
Aortic	50-65 y	Mechanical OR bioprosthetic (2a)
Aortic	> 65 y	Bioprosthetic AVR (2a)
Mitral (if repair not possible)	< 65 y	Mechanical MVR (2a)
Mitral	≥ 65 y	Bioprosthetic MVR (2a)

- **VKA contraindicated, cannot be managed or not desired** → ★ **bioprosthetic valve (class 1)**.



Slide 58: prosthetic valve selection (ACC/AHA 2020)

	Mechanical [added]	Bioprosthetic [added]
Durability	Lifelong	10-15 years (faster degeneration in the young)
Anticoagulation	Lifelong warfarin (INR 2.5-3.5 depending on position/risk); DOACs contraindicated	Usually 3-6 months, then aspirin
Best for	Young, can take warfarin	Elderly, women wanting pregnancy , bleeding risk, poor adherence

Thromboembolic risk factors for prosthetic valves (slide 59):

- **Older-generation valve** (e.g., caged-ball)
- **AF**
- **Previous thromboembolism**
- **Hypercoagulable state**
- **LV systolic dysfunction**

🔄 **Recap: Prosthetics.** Mechanical valves last but need lifelong warfarin; bioprosthetic valves need no long-term VKA but wear out. Aortic: < 50 mechanical (or Ross), 50-65 either, > 65 bio. Mitral: < 65 mechanical, ≥ 65 bio. Can't take VKA → bio. Higher thrombosis risk with older valves, AF, prior emboli, hypercoagulability, poor LV. All prosthetic valves are **high risk for IE** (amoxicillin before dental extraction).

3. High-Yield Comparison Tables

Table A: The Big Five Lesions

	AS	AR	MS	MR	TR
Commonest cause	Calcific (old), bicuspid (young)	Bicuspid (rich countries), rheumatic (poor); root disease	Rheumatic	MVP (developed), ischaemic, DCM	Functional (RV dilatation)
Load	Pressure (LV)	Volume (LV)	Pressure (LA)	Volume (LA + LV)	Volume (RA/RV)
Murmur	Ejection crescendo-decrescendo , R 2nd ICS → carotids	Early diastolic decrescendo , LSE, leaning forward	Mid-diastolic rumble , apex, opening snap	Holosystolic , apex → axilla	Holosystolic , LLSE → apex, ↑ inspiration
Heart sounds	Soft S2, S4 , ejection click	S3 , Austin Flint	Loud S1 , OS	Soft S1, S3	
Pulse / periphery	Parvus et tardus , narrow PP	Collapsing , wide PP, Hill	AF, malar flush	± AF	Giant v waves, pulsatile liver
ECG / CXR	LVH, valve calcium	LV dilatation, prominent root	P mitrale , AF , big LA	LVH, P mitrale, big LA + LV	RA/RV enlargement
Key symptom	SAD	Palpitations, dyspnoea	Dyspnoea, haemoptysis , hoarseness , dysphagia	Dyspnoea, fatigue	Ascites, oedema
Intervention	AVR (SAVR/TAVR) if symptomatic, ↓EF, or other cardiac surgery	AVR if symptomatic, EF ≤ 55% , LVESD > 50	PMBC first	Repair if symptomatic or EF ≤ 60% ; acute → urgent	Treat the cause <i>[added]</i>

Table B: Dynamic Auscultation (Manoeuvres)

Manoeuvre	Effect on haemodynamics	AS	MR	AR	MVP	HCM <i>[added]</i>
Handgrip	↑ afterload	↓	↑	↑	↓ (later click)	↓
Valsalva / standing	↓ preload	↓ or =	↓	↓	↑ (earlier click, longer)	↑
Squatting / leg raise	↑ preload	↑	↑	↑	↓	↓
Inspiration	↑ right venous return	—	—	—	—	— (TR ↑)

Table C: Acute vs Chronic Regurgitation

	Acute AR / MR	Chronic AR / MR
Causes	IE, dissection, trauma (AR); papillary rupture, chordal rupture, IE (MR)	Valve or root disease, MVP, RF, DCM
LV	Normal size (no time to adapt) → pressures spike	Dilated , compliant
Presentation	Sudden pulmonary oedema, shock	Silent for years, then HF

	Acute AR / MR	Chronic AR / MR
Signs	Murmur may be short or soft; no wide pulse pressure (AR)	Classic murmurs + peripheral signs
Rx	★ Urgent surgery + afterload reduction	Surgery by symptoms and EF/LV size

Table D: Surgical Thresholds from This Lecture

Lesion	Operate when
AS	Symptomatic · asymptomatic severe with ↓ LVEF · undergoing other cardiac surgery
AR	Acute severe · symptomatic chronic severe · asymptomatic severe with LVEF ≤ 55% (consider if LVEDD > 50 mm)
MR	Acute primary → urgent · chronic severe primary: symptomatic, or asymptomatic with LVEF ≤ 60%
MS	Severe → PMBC preferred; else surgery
MVP	Usually none

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Murmurs louder with ↑ preload	MVP and HCM get louder/longer with ↓ preload (Valsalva, standing)
Handgrip ↑ regurgitant murmurs	It ↓ AS and HCM <i>[added]</i>
Left-sided murmurs ↑ on expiration	Right-sided (TR) ↑ on inspiration
AS = elderly	Bicuspid valves give AS (and AR) in younger adults
Regurgitation → dilatation	Acute regurgitation has a normal-sized LV , hence sudden oedema without peripheral signs
MR EF "normal" = 55–70%	In MR, EF ≤ 60% already means dysfunction
Valve replacement for MR	Repair is preferred
MS → surgery	PMBC is first choice if anatomy suits
TR = valve disease	Usually functional (RV dilatation)
Mechanical valve for the young	Bioprosthetic if VKA can't be used (incl. pregnancy wishes <i>[added]</i>)
Ascites in HF follows leg oedema	In TR (and constriction), ascites precedes LL oedema

5. Clinical Correlations

- **Elderly patient with exertional syncope, angina, slow-rising pulse → severe AS** → echo → TAVR/SAVR. Avoid nitrates.
- **Tall young man with long limbs, lens dislocation and an early diastolic murmur → Marfan root dilatation → AR** (also MVP). *[added]* Watch for dissection.
- **Sudden pulmonary oedema 5 days after inferior MI with a new apical pansystolic murmur → papillary muscle rupture → acute MR** → urgent surgery. *[CAD lecture link]*
- **Fever + new early diastolic murmur + heart block → IE with aortic root abscess** → surgery. *[IE lecture link]*

- **Tearing chest pain + new AR murmur → type A dissection** → emergency surgery.
- **Young woman from a low-income country, pregnant, with dyspnoea, haemoptysis, AF → rheumatic MS** (pregnancy volume load unmasks it). *[added]* β -blocker $\pm$ PMBC.
- **Hoarseness + loud S1 + opening snap → Ortner syndrome** from a giant LA in MS.
- **IVDU with pansystolic murmur ↑ on inspiration and a pulsatile liver → TR** from tricuspid IE.
- **55-year-old needing AVR who refuses warfarin → bioprosthetic** valve.
- **Mechanical valve + AF + previous stroke** → highest thromboembolic risk → strict INR, *[added]* $\pm$ aspirin.

6. Rapid Revision (Last-Day)

1. Stenosis = pressure overload → hypertrophy; regurgitation = volume overload → dilatation; both → HF.
2. **AS** is the commonest valve lesion in industrialised countries (degenerative, > 75; bicuspid if young). **MR** is second. **MS** presents at 20–39.
3. **Acute AR:** IE, dissection, trauma → sudden oedema. **Chronic AR:** bicuspid (rich), rheumatic (poor), root dilatation (Marfan, EDS, syphilis).
4. AR signs: water-hammer/Corrigan, **De Musset, Becker, Müller, Quincke, Traube, Duroziez, Hill (≥ 60 mmHg, most sensitive)**.
5. AR murmur: **early diastolic decrescendo**, LSE, leaning forward; **S3; Austin Flint**.
6. AR surgery: acute severe · symptomatic · asymptomatic with **EF $\leq 55\%$** (consider **LVESD > 50 mm**).
7. AS causes: **calcific (commonest), bicuspid (earlier)**, rheumatic. **SAD. Pulsus parvus et tardus**, narrow PP.
8. AS murmur: **diamond ejection → carotids**; ↓ with handgrip and Valsalva/standing; **soft S2, S4, ejection click**.
9. AS: echo first; cath for exact area/gradient. **AVR (SAVR/TAVR)** if symptomatic, ↓ EF, or other cardiac surgery; balloon valvuloplasty is an option.
10. **MR primary:** MVP, annular calcification, ruptured chordae, RF, IE. **Secondary:** ischaemic papillary dysfunction, DCM.
11. MR: **holosystolic, apex → axilla**, ↑ handgrip and leg raise; S3; **P mitrale**; LA + LV enlargement.
12. **Acute primary MR → urgent surgery; repair preferred**. Chronic severe: symptomatic or **EF $\leq 60\%$** .
13. **MVP:** commonest MR cause in developed countries; myxomatous; **mid-systolic click + late murmur; squatting ↓**; usually no treatment; IE risk.
14. **MS:** rheumatic; dyspnoea, **hoarseness, dysphagia**, later right HF, PND, orthopnoea, **haemoptysis; malar flush**.
15. MS: **loud S1, opening snap, apical diastolic rumble**; P mitrale; big LA / LA appendage on CXR; **PMBC preferred**; complications AF, oedema, pulmonary HTN, CHF.
16. **TR:** mostly functional; congested pulsating neck veins, pulsatile tender liver, **ascites before oedema**; pansystolic murmur **↑ on inspiration**, to the apex **not the axilla**.
17. **Prosthetics:** aortic < 50 mechanical (or Ross), 50–65 either, > 65 bio. Mitral < 65 mechanical, ≥ 65 bio. **No VKA → bio**.
18. Prosthetic thrombosis risk: **old-generation valve, AF, previous TE, hypercoagulable state, LV dysfunction**.

7. Exam-Style Questions

Q1 (lecture case). A 75-year-old woman has exertional syncope, 3 years of exertional dyspnoea and chest discomfort, and weak delayed radial pulses. On the LV pressure-volume vs time plot (points A–E), where does her murmur start? A · **B** · C · D · E

Answer: B. *[reasoning added; key not on slides]*

- **SAD + pulsus parvus et tardus = aortic stenosis.** Its murmur is an **ejection** murmur that starts when the **aortic valve opens**.
- That is point **B**, where LV pressure has risen and **LV volume starts to fall**.
- **A** = mitral closure (S1, isovolumetric contraction; no flow across the aortic valve yet).
- **C** = aortic closure (S2; an AR murmur would start here).
- **D** = mitral opening (an MS rumble/opening snap would start here).
- **E** = late diastole.

Q2. A 30-year-old man with Marfan syndrome has a high-pitched early diastolic decrescendo murmur at the left sternal edge, BP 150/50, and head bobbing. Echo: severe AR, LVEF 52%, LVESD 48 mm. He is asymptomatic. What is the management? A. Observe, repeat echo in 1 year · B. **Aortic valve (± root) replacement** · C. Start β -blocker only · D. Balloon valvuloplasty · E. TAVR

Answer: B.

- **Asymptomatic severe AR with LVEF $\leq$ 55%** is a surgical indication (slide 23).
- *[added]* In Marfan, the root is usually replaced too.
- Balloon valvuloplasty and TAVR are for **stenosis** (TAVR is not standard for AR).

Q3. Six days after an inferior STEMI, a patient develops sudden severe dyspnoea, hypotension and a new apical holosystolic murmur radiating to the axilla. Echo shows a flail posterior leaflet. What is the definitive management? A. IV furosemide and observe · B. **Urgent mitral valve surgery (repair if possible)** · C. Elective MitraClip in 3 months · D. PMBC · E. Pericardiocentesis

Answer: B.

- **Acute primary MR** (papillary muscle rupture) → **urgent surgical repair or replacement for all**, with medical stabilisation meanwhile (slide 39).
- PMBC is for **MS**.

Q4. Which manoeuvre makes the murmur of mitral valve prolapse **softer and shorter**, with the click moving later? A. Valsalva · B. Standing from squatting · C. **Squatting** · D. Amyl nitrite · E. Inspiration

Answer: C.

- **Squatting $\uparrow$ preload** → LV larger → less prolapse → later click and shorter murmur (slide 44).

Q5. A 52-year-old needs aortic valve replacement and has no contraindication to warfarin. Which ACC/AHA 2020 option is reasonable? A. Mechanical only · B. Bioprosthetic only · C. **Mechanical OR bioprosthetic after shared decision-making** · D. Ross procedure mandatory · E. Balloon valvuloplasty

Answer: C.

- **Age 50-65 → either valve (2a).** Mechanical (2a) is for < 50; bioprosthetic (2a) for > 65 (slide 58).

8. Exam Pitfalls & Traps

- **Slide 21 (AR diagnostics)** says TTE is used "to confirm diagnosis and determine **AS** severity". This is a copy error from the AS slides; it means **AR** severity.
- **Hill sign:** the slide gives ≥ 60 mmHg and calls it the "most sensitive sign for AR". Classically > 20 mmHg is abnormal and **> 60 mmHg indicates severe AR** [added]. Quote ≥ 60 mmHg for the exam.
- **AS manoeuvres:** handgrip **decreases** AS, Valsalva decreases or doesn't change it. **Only HCM (and MVP) get louder with Valsalva/standing**. This is the classic AS vs HCM discriminator [added].
- **EF cut-offs differ: AR $\leq 55\%$, MR $\leq 60\%$.** Don't mix them.
- **MR "repair preferred":** replacement is a wrong answer when repair is feasible.
- **MS intervention: PMBC first**, not valve replacement, if anatomy suits.
- **Acute regurgitation** lacks the classic chronic signs. A short or soft murmur with flash pulmonary oedema is still acute MR/AR.
- **TR radiates to the apex but NOT the axilla.** MR radiates **to the axilla**. Inspiration $\uparrow$ TR only.
- **Ascites before leg oedema** points to **TR or constrictive pericarditis**, not left HF.
- **Malar flush (MS)** vs **butterfly rash (SLE)**: the SLE rash spares the nasolabial folds and is photosensitive.
- **Prosthetic valve selection** follows **ACC/AHA 2020 age bands** (slide 58). ESC uses slightly different ages [added]. Use the slide's numbers.
- **DOACs are contraindicated with mechanical valves** [added]. Warfarin only.
- **Slide 57 valve letters (A-H) are not labelled on the slide.** The identification above is reconstructed; know the categories (mechanical: bileaflet, tilting disc, caged-ball; bioprosthetic: stented, stentless; transcatheter) rather than the letters.